

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Architected a full-stack **ML training-data** annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Implemented backend security and reliability controls, AWS KMS encryption, auth hardening, rate limiting, and integration tests, remediating **20+ VAPT findings**.
- Served as primary technical POC and piloted a COO-sponsored GitHub Copilot experiment, onboarding a second developer to on-call readiness.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Shipped a production LLM feature (auto-commenting via **4-bit quantized Llama-7B, llama.cpp**) with per-persona prompts, async queued generation, and a content-safety filter, built end to end with no ML mentor.
- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) and migrated core repositories from JavaScript to TypeScript.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform independently end to end; live **3+ years**.

Projects

Refusal Decay – LLM Safety / Mechanistic Interpretability (GitHub) PyTorch, Llama-3.1-8B, AdvBench

- Ran a mechanistic-interpretability study of how prefilling attacks weaken refusal in Llama-3.1-8B-Instruct, extracting the refusal direction via difference-in-means (refusal **0.92**→**0.32** at $k=3$) with causal patching, bootstrap 95% CIs, and McNemar's test.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, PyTorch, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided **RAG** pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a **4,905-question** evaluation.

ngvi-curvature-variance – Natural-Gradient VI Study (GitHub) Python, PyTorch, NumPy

- Ran a matched-compute study of when natural-gradient VI beats Adam on ill-conditioned posteriors (**PyTorch**, black-box VI), showing NGVI's ELBO gain grows with curvature (κ 180 to 2566 maps to **+0.28 to +2.32 nats**).

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, Bash

Deep Learning: PyTorch, JAX, Hugging Face Transformers, NumPy, transformers, normalizing flows, DDPM

NLP & LLMs: NLP, LLM evaluation, mechanistic interpretability, RAG, retrieval, prompt engineering

Methods & Stats: variational inference, natural-gradient optimization, autodiff from scratch, bootstrap CIs, McNemar's test

Engineering & Cloud: React, Next.js, FastAPI, Docker, Git, AWS, GCP, Azure, Linux

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.